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K-Means Porosity Clustering in MRI-Scanned Reef Rock: Insights from Zechstein Limestone, Poland

A. Fheed, Department of Energy Resources, AGH University of Krakow, Kraków, Lesser Poland, Poland; R. S. Babu, A. R. Adebayo, and A. El-Husseiny, Center for Integrative Petroleum Research, King Fahd University of Petroleum and Minerals, Dhahran, Eastern Province, Saudi Arabia

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Abstract

Objectives/scope: The study aimed to segment the pore space in a 3D MRI scan of a reef limestone using k-means clustering. Voxels with similar proton densities were grouped into 3D clusters. Traditional MRI interpretation struggles with highly heterogeneous samples, as automatic color mapping can create an excessive number of value ranges, reducing the readability of an image. Unlike manual segmentation, this workflow simplifies the image while retaining essential structural details of the pore space.

Methods, procedures, process: A brine-saturated bioclastic limestone sample from the Zechstein Limestone Formation (Ca1, Poland) was subjected to MRI scanning at 30 MHz. The 3D SE-SPI sequence with echo time of 2000 μ s, repetition time of 10000 ms and voxel size of 0.625 mm was used. The image consisting of mL-scaled voxels was segmented using k-means clustering. The number of clusters was determined using the elbow method, resulting in four clusters. The first proton density cluster containing noise was excluded. The remaining three clusters had their volumes computed along with the mean, standard deviation and median of the voxel values they contained.

Results, observations, conclusions: The analysis showed that the three identified pore domains characterized by low, moderate and high proton density, had volumes ranging from around 0.7 mL in the low and medium proton density clusters to roughly 0.4 mL in the high proton density cluster. The mean voxel values showed an increase of at least 20% between successive clusters of growing proton density, while the standard deviation was the highest in the cluster with the greatest proton density. The segmented image revealed multiple bendy, channel-shaped dissolution features, containing varying amounts of water. In places where dissolution had intensified, larger pore domains were created. The k-means clustering proved to be a robust method for distinguishing similar pore groups in MRI data, allowing for their consistent and reproducible quantification.

Novel/additive information: Compared to medical studies, the k-means MRI data clustering remains uncommon in petrophysics. We facilitate MRI-based pore structure analysis in carbonate rocks by offering

a data-driven alternative to manual segmentation. This unbiased approach enables consistent classification of porosity. Unlike traditional color mapping, which can obscure critical details in complex pore networks, clustering highlights meaningful variations in proton density. The proposed methodology enhances porosity assessment and provides insights into permeability by identifying interconnected pore pathways.

Introduction

Carbonate reservoirs have long been a common target for the petroleum industry as many of them host valuable hydrocarbon resources (Garland et al., 2020) and nowadays are gaining popularity for geothermal exploration (Wadas et al., 2023) and underground gas storage, including CO₂ sequestration (e.g. Moghanloo et al., 2017; Seyyedi et al., 2020; Razi-perchikolaee et al., 2021). Reef carbonates are a specific subgroup of carbonate rocks which is distinct in terms of their petrophysical parameters. These rocks tend to be very heterogeneous, which stems from their complex depositional settings, including biological construction, episodic sediment infill, early cementation and secondary porosity development during diagenesis (Flügel, 2010; Hollis et al., 2010; Fheed et al., 2022). As a result, their pore systems exhibit a wide range of shapes, sizes and connectivity patterns (Strzelecki et al., 2020). Capturing this complexity requires methods that preserve the spatial variation in fluid distribution, rather than simplifying it to a simple pore-matrix view. In times of increased demand for energy and growing environmental awareness, the relatively understudied reef rocks therefore seem to be an interesting research target.

Despite their petrophysical potential, related studies on reef carbonates remain scarce. This concerns the two commonly used imaging techniques revealing their pore space architecture: (1) X-ray microtomography (μ CT) and (2) Magnetic Resonance Imaging (MRI). This study explores the use of unsupervised clustering as an image post-processing tool for MRI data analysis in reef carbonates. We aim to provide a scalable and reproducible workflow that enhances visualization and facilitates petrophysical interpretations. Using such a technique can improve our ability to assess reservoir quality, assist in upscaling petrophysical properties for modeling purposes and identify promising reservoir zones for hydrocarbon recovery or CO₂ injection. Although it is undeniable that μ CT imaging usually provides more detailed results of better resolution than the MRI does, the method has a downside of interpreter bias while choosing the proper threshold between the rock matrix and pores which is not the case for MRI which only represents fluid-filled areas (Lev and Boyce, 2020; Krzyżak et al., 2022).

Raw MRI data usually need some post-processing to improve their clarity and extract more information about the sample (Karlsens et al., 2021). Filtering can significantly increase the readability of the images as keeping the entire signal range in the image makes detailed analysis challenging, specifically in regions containing smaller pores. It also allows for distinguishing between signal components of different proton density (PD) to gain better visualization control on them, which would be highly constrained without this step. Manual filtering is not an optimal solution due to its highly subjective character while automatic color coding of the original images can result in exaggerated number of classes making them difficult to read. Therefore, the goal of this paper is to present an easy way of achieving unbiased magnetic resonance image categorization for heterogeneous reservoir rocks such as reef rocks, supporting the evaluation of their quality for the needs of hydrocarbon exploration or CO₂ sequestration.

Material and methods

Sample selection and characterization

One sample of relatively complex pore network was selected for the study. The sample was a bioclastic limestone containing dissolved shells of foraminifera which have formed a bendy pore space noticeable in microscope images. Microscope studies were conducted in transmitted light on one thin section sample stained with blue epoxy facilitating porosity identification. The sample comes from the Zechstein Limestone

Formation (Ca1) of west Poland and was withdrawn from the Brońsko Reef, being a major reef body consisting of various limestone to dolomite rock types. It had a cylindrical shape, the diameter of 1 in. (2.54 cm) and height of approximately 1.54 in. (3.91 cm).

Sample saturation and MRI experiments

To saturate the sample with a hydrogen-bearing fluid, it was first placed in a vacuum chamber and then filled with 3% NaCl artificial brine under the pressure of 2000 PSI for several hours to ensure its full saturation. The sample was then wrapped with plastic foil to prevent desaturation. A multifrequency MRI equipment set to magnetic field of 30 MHz was used in the experiment. 3D SE-SPI imaging sequence was deployed. The echo time (TE) was set to 2000 μ s whereas the repetition time (RT) equaled 10000 ms. The resolution of the method was $128 \times 128 \times 128$ with the field of view (FOV) of 3.15 in. (80 mm), giving the voxel size of roughly 0.025 in. (0.625 mm).

Image post-processing

Exporting and cropping the image. The original image was obtained in the Green Imaging Technologies (GIT) software, where it was scaled to mL. After exporting the 3D image, it was loaded to Fiji platform, where it was preliminarily cropped by eliminating the empty probe space around the sample. The image was then exported back as a 3D TIFF and imported to MATLAB.

Clustering. The clustering procedure was conducted using MATLAB. Before continuing, a cylindrical mesh grid was created to eliminate the background left after the rectangular cropping in Fiji. The mesh grid was extended to 3D and underwent a 3D gaussian filtering to make it smoother. The mask was applied to the 3D image which was binarized. All points outside the cylinder were assigned the value of 0, whereas the remaining points were assigned the value of 1. After excluding the surrounding of the sample, the k-means clustering was performed.

The number of clusters was chosen using the elbow method. K-means clustering was applied to the MRI data for values of k from 1 to 10. For each value, the total within-cluster variation (inertia), which represents how close the data points are to the centers of their assigned clusters, was calculated and plotted (Fig. 1). The optimal number of clusters was determined by identifying the point where the curve began to flatten, indicating that 4 clusters are optimal in our case.

Subsequently, the clustering aiming to subdivide the magnetic resonance image into several regions of different PD began with the constraint of 4 clusters. Each voxel had specific value assigned based on the clustering results. Because non-porosity voxels were still present within the cylindrical region, the first cluster representing extremely low intensity components was excluded from further analysis. For the remaining three clusters, total volumes, number of voxels and voxel intensity statistics including minimum, maximum, mean, median and standard deviation values; were computed.

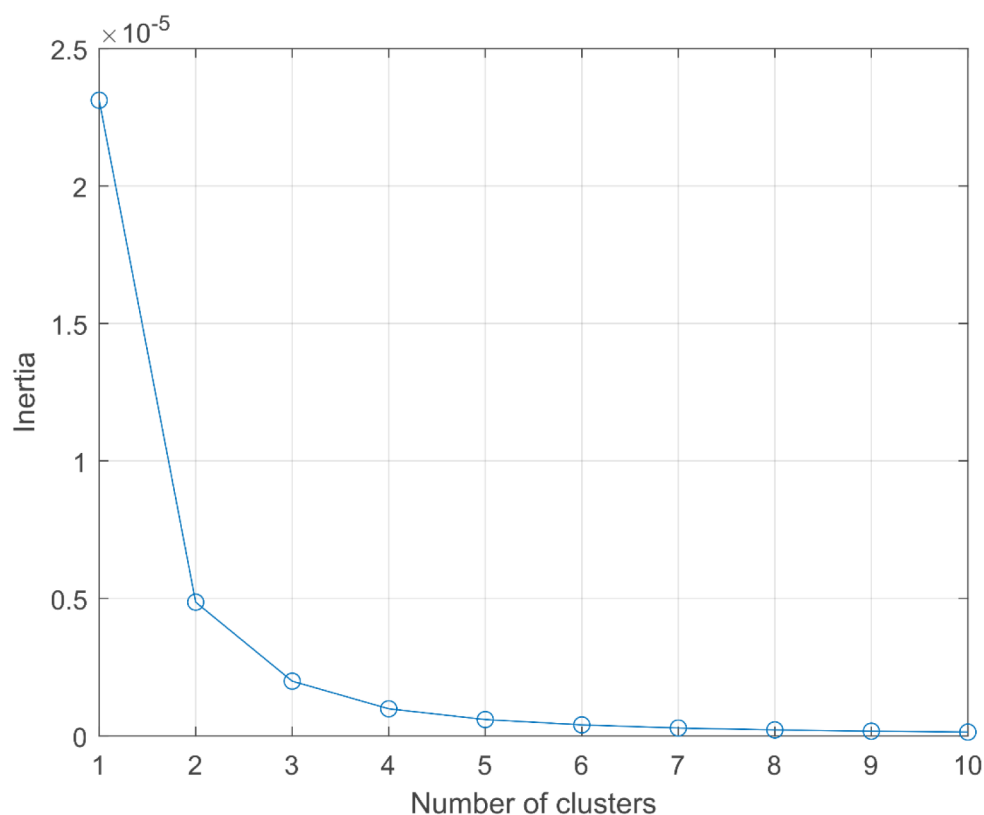


Figure 1—The elbow method for determining the number of clusters. Inertia values are measured in mL^2 , reflecting the sum of squared distances between the voxel values and their respective centroids. Curve flattening can be noticed around 4 clusters, and therefore this number was found optimal for the analysis.

Results

Microscope studies

Microscope research revealed sample characteristics (Fig. 2). It can be classified as a bioclastic packstone or boundstone with bryozoans and foraminifera. It comprises numerous moldic pores formed after the dissolution of the foraminifera shells. The pore system is dense and bendy, while in many places the molds had been merged to form vuggy porosity. The framework of the rock is currently represented mainly by drusy calcite mosaic.

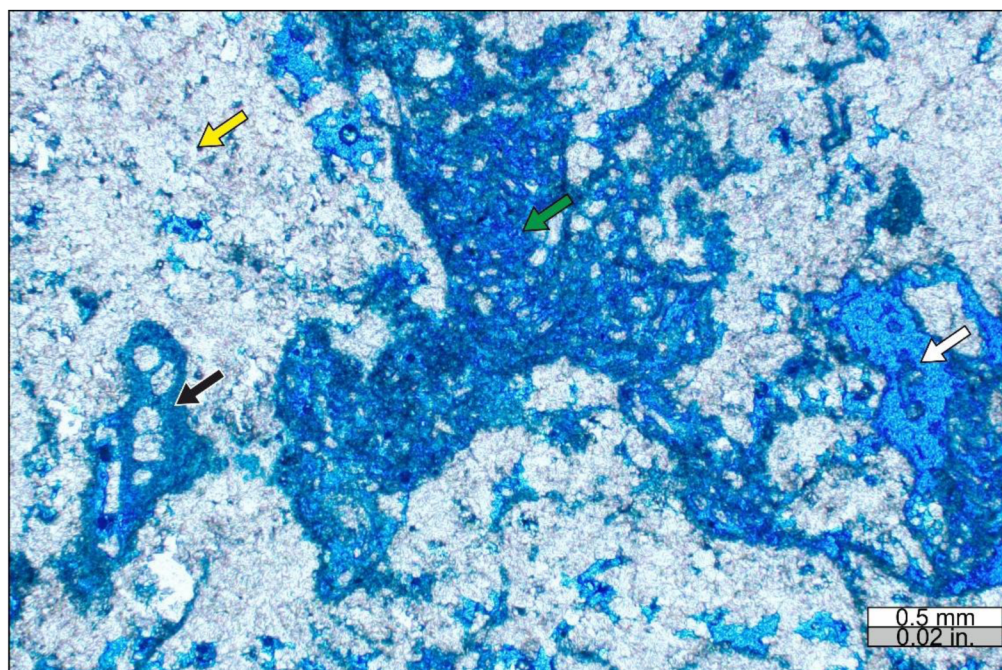


Figure 2—Transmitted-light microphotograph of the sample taken at 1N. Blue stain marks porosity. Description of arrows: black – partially dissolved bryozoan fragments; green – molds after dissolved shells of encrusting foraminifera forming bendy pore networks; white – vuggy pores created in places where dissolution had intensified, and individual molds were merged; yellow – drusy calcite cement.

MRI data processing results

The three obtained image clusters were classified as low, moderate and high PD regions. Fig. 3 shows 3D visualization of those regions while Table 1 presents quantitative results related to the corresponding clusters. The high PD cluster had observably lower volume compared to the low and medium PD ones: 0.39 mL compared to approximately 0.7 mL in case of the remaining two.

Table 1—Basic statistics for the three analyzed MRI clusters. All parameters except for the number of voxels are scaled in mL.

Parameter	Low proton density cluster	Medium proton density cluster	High proton density cluster
Min	$3.00 \cdot 10^{-5}$	$3.77 \cdot 10^{-5}$	$4.66 \cdot 10^{-5}$
Max	$3.77 \cdot 10^{-5}$	$4.66 \cdot 10^{-5}$	$7.28 \cdot 10^{-5}$
Median	$3.44 \cdot 10^{-5}$	$4.09 \cdot 10^{-5}$	$5.13 \cdot 10^{-5}$
Mean	$3.42 \cdot 10^{-5}$	$4.13 \cdot 10^{-5}$	$5.20 \cdot 10^{-5}$
Standard deviation	$2.14 \cdot 10^{-6}$	$2.45 \cdot 10^{-6}$	$3.90 \cdot 10^{-6}$
Cluster volume	0.70	0.66	0.39
Number of voxels	20384	15959	7553

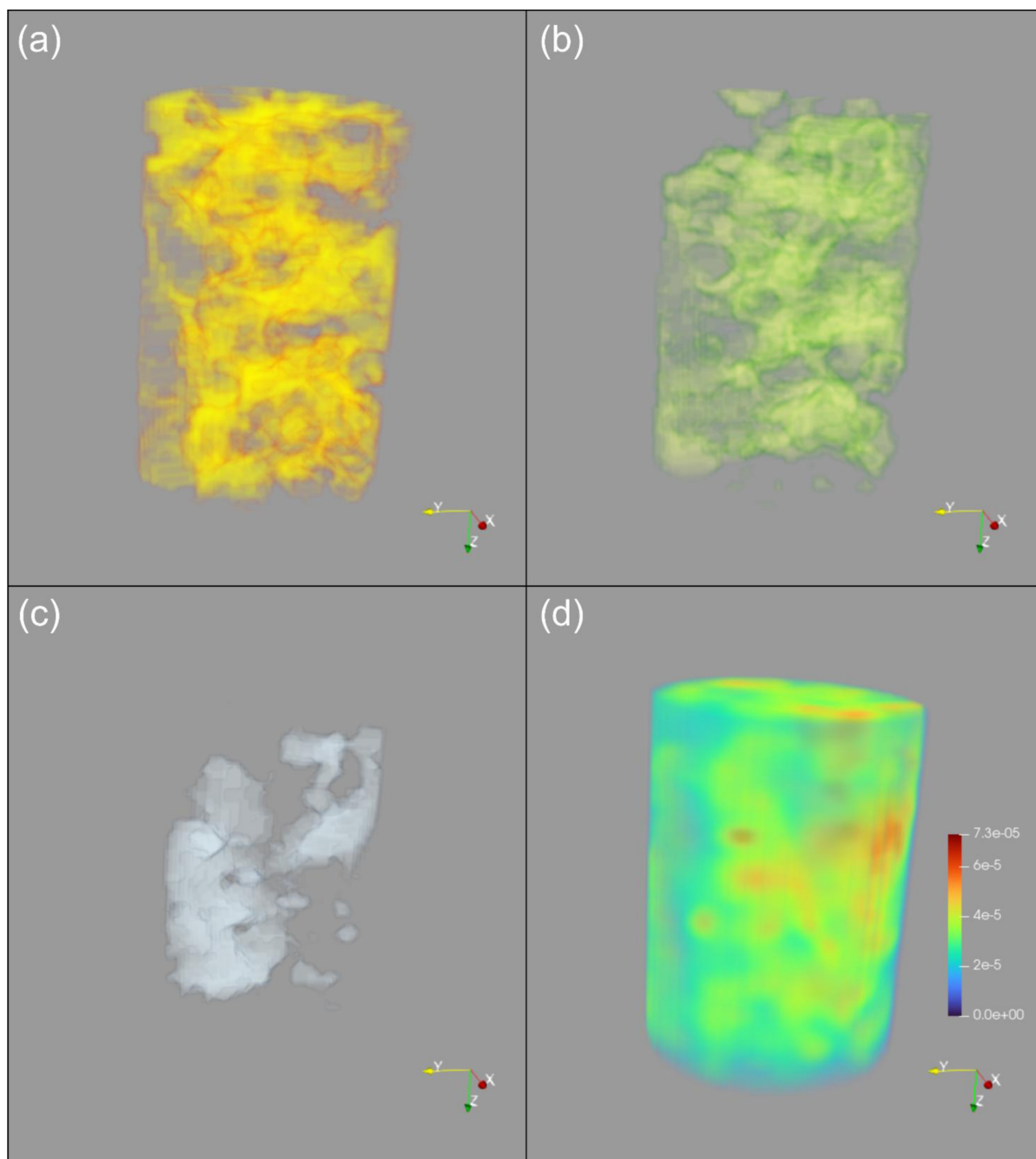


Figure 3—Clustering results: (a) low proton density region; (b) moderate proton density region; (c) high proton density region; (d) whole sample image with a color scale bar indicating the amount of brine per voxel scaled in mL. All images have the same orientation with x, y and z directions marked, where z is the longest axis of the plug. Note that the outlines of the high PD cluster can be easily traced in picture (d), whereas it is challenging to clearly see the outlines of the remaining clusters in the original image.

Qualitative assessment of the obtained images gave insights into the pore network details. Numerous bendy channel-like areas were noted and most of them belonged to either low or medium PD clusters (Fig. 3). Similar outlines of the apparent geological structures occur in the two mentioned clusters (Fig. 3). The high PD cluster looks differently, opting to the occurrence of larger structures resembling dissolution vugs (compare Figs. 2 and 3). Although representing the smallest volume (0.39 mL) and voxel count (7553; see

Table 1), it exhibits the highest mean voxel signal intensity value ($5.20 \cdot 10^{-5}$ mL) and variability of data expressed by the standard deviation of $3.90 \cdot 10^{-6}$ mL. In contrast, the low and medium PD clusters cover larger volumes of 0.7 mL and 0.66 mL, respectively, but have lower corresponding mean values of $3.42 \cdot 10^{-5}$ mL and $4.13 \cdot 10^{-5}$ mL. The median values in all cases stay close to the average. The sum of all cluster volumes gives approximately 1.75 mL. Considering the volume of the analyzed region after cropping, being roughly 14.17 mL, the resulting bulk porosity is 12.75%. This value is expected to slightly differ from the standard low-field (LF) NMR method due to the following reasons: (1) the LF NMR normally considers the entire volume of the sample; (2) the MRI data is collected at 30 MHz compared to much lower field typically close to 2 MHz for the LF devices that could potentially generate some internal magnetic field gradients and introduce some signal attenuation; (3) small portion of signal might be deleted as part of the first cluster when eliminating voxels representing noise; (4) the echo time used for SE-SPI sequence is often longer than the corresponding LF value used in a standard Carr-Purcell-Meiboom-Gill (CPMG) sequence. Nevertheless, this paper did not focus on porosity computation but rather on MRI data categorization and visualization, but it is worth mentioning that the differences between the two methods should be relatively small so that porosity can also be obtained that way.

Discussion

K-means clustering

The k-means clustering method is well-known in MRI data processing. However, in petrophysical applications it is not as popular as in medical studies (e.g. [Bellezza et al., 2024](#)). In geology, k-means clustering is sometimes combined with machine learning to extract information useful in mineral exploration, but this application still falls beyond the scope of MRI (e.g. [Liu et al., 2019](#); [Wróblewska-Janc, 2024](#)). This is likely due to the fundamentally different contrast characteristics of medical and petrophysical data, which reduces the relevance of classifying rock images in the same way as in medical imaging. Nevertheless, for highly heterogeneous rocks, clustering techniques, such as k-means, provide a systematic, data-driven way to categorize MRI signal into meaningful proton density classes. These techniques can reduce interpreter bias and overcome the limitations of manual filtering or arbitrary color mapping. Instead of relying on visual intuition alone, clustering algorithms segment the image based on underlying signal distributions, making it easier to identify zones of different pore space characteristics. This can be particularly advantageous in characterizing the internal architecture of reef reservoirs, where traditional segmentation fails to emphasize pore regions with moderate proton density. By following the presented approach, it is possible to select and extract clearly defined porosity regions characterized by similar PD levels that are no longer hidden behind the dense pore network.

The effectiveness of clustering, however, strongly depends on the image quality and resolution. In the presented study, a relatively high-resolution MRI scan made it possible to capture internal rock heterogeneity in detail. Still, some finer features, especially narrow pore throats, may not be fully resolved (see [Lev and Boyce, 2020](#)), potentially leading to underrepresentation of the connected regions in the final segmentation outcome.

MRI and PD weighting in reef carbonates

Reef carbonates are a good match for MRI studies because their low ferromagnetic mineral content leads to minimal internal magnetic field gradients, even at high magnetic field strengths (see [Fheed et al., 2018](#)). In such cases, signal intensity will normally be good enough to consider the 3D imaging even at sub-millimeter resolution. This of course does not apply for some tighter facies surrounding the reefs and lithologies such as mudstone or wackestone, for which the TE values used in the SE or SE-SPI sequences are too long to record meaningful signals and SPRITE imaging could be the only option (see [Zamiri et al., 2022](#)). Considering worse SNR for the last mentioned, the SE-SPI sequence was found optimal in our case.

The contrast in MRI can be controlled by manipulating basic acquisition parameters. Thus, the image can be T1, T2 or proton density-weighted. The last option is typically chosen when identifying regions representing various amounts of hydrogen protons, pointing to different saturation levels (Marica et al., 2006; Li et al., 2009). Since PD weighting was our aim, the chosen RT was long, whereas the TE was kept relatively short. Using a PD-weighted image made it possible to deduce porosity evolution in the studied plug.

Geological interpretation of the obtained results

Multiple dissolution features could be observed in the image, forming variously oriented paths. In the low and medium PD clusters, the results suggest the occurrence of uniformly distributed pore regions, likely consisting of voids having limited sizes compared to those from the high PD cluster, confirmed by relatively low mean signal intensity values. The narrow standard deviations in these clusters imply quite homogeneous porosity domains. By contrast, much higher mean value of the signal intensity and higher standard deviation for the high PD cluster increases the likelihood of finding larger and more heterogeneous pore bodies there. They are probably associated with partially or fully merged bioclast-related dissolution features, forming a connected macropore network. Such a phenomenon seems very realistic in natural porous media, and especially in reef carbonates, largely composed of easily soluble bioclasts often exposed to aggressive fluids (Scholle and Ulmer-Scholle, 2003; Flügel, 2010). In this case, supporting analyzes point to the presence of multiple dissolved foraminifera shells in the rock, clarifying the main way of porosity creation. The studied sample has a complex nature. Although it has a consistently developed pore network, specifically in both low and moderate PD clusters, it still comprises local heterogeneities, especially noticeable in the high PD cluster. The merged pore networks might be a decisive factor when controlling local permeability of the reservoir, but further work is needed to make more conclusions regarding this matter.

Potential applications of the method to CO₂ storage

Finally, in relation to underground CO₂ storage activities, there is still not a clear consensus on whether permeability increases or decreases after the contact with supercritical CO₂ (Silveira et al., 2022). Hopefully, the use of MRI data clustering could be a step forward in tracing the potential changes to rock structure before and after the flooding and/or ageing of a rock sample as a solid support for μ CT imaging. Some sources point to debris accumulation after the initiation of dissolution that could form obstacles for the effortless fluid flow through a rock (Silveira et al., 2022). Such zones could likely be spotted in the magnetic resonance images as extremely low PD regions. However, we leave this as a potential future application since this paper focuses solely on one sample unaffected by CO₂ and its principal aim was to test the effectiveness of the clustering for an improvement of MRI data visualization.

Conclusions

K-means clustering seems to be an efficient tool in rock MRI data segmentation and categorization. If image categorization is neglected, the voxels associated with areas likely linked to the occurrence of structures containing smaller amounts of fluids may appear very faint or even not be visible at all using standard color coding of the entire image. The k-means clustering proved to be an appropriate solution for distinguishing MRI regions represented by similar PD, thus enhancing the robustness of their visualization.

Although MRI data cannot be directly associated with geological structures due to resolution limitations, it seems that for highly porous, conventional rocks such as reef carbonates, the object outlines are relatively well-exposed after the clustering procedure. The presented method allowed for precise examination of the pore space and had an advantage over the μ CT data in terms of its unbiased character. Moreover, large amounts of elongated pore spaces make it difficult to extract them from the μ CT image without the risk of including non-porosity components during the thresholding.

In the studied case, the applied methodology helped recognize the main driving factor behind the origin of secondary pores in the analyzed reef rock sample. The expected process responsible for the creation of most of the pore spaces was likely the dissolution of easily soluble foraminifera shells often attached to or surrounding the bryozoan fragments. Each time the dissolution had been strong enough to force merging of individual, channel-like structures, vuggy porosity was being formed, ultimately enhancing petrophysical parameters of the sample. On the reservoir scale, this phenomenon could have a decisive role in permeability enhancement, which might be of particular interest when searching for optimal gas storage sites or conducting hydrocarbon exploration tasks. Future research will focus on analyzing changes to reef rock texture after the CO₂ flooding and ageing, where MRI could be supportive for μ CT data analysis.

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